

Reason about rounding boundaries

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: What is the smallest whole number that rounds to 86,000 to the nearest 1,000? Copy or complete the first step.

2. A number rounds to 4,300,000 to the nearest 100,000. What is the greatest possible number?

3. Is 2,650,000 rounded to the nearest million 3,000,000?

4. Miro says: Miro includes 85,499 in the range that rounds to 86,000. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: What is the smallest whole number that rounds to 86,000 to the nearest 1,000? Copy or complete the first step.
Answer 85,500.
2. A number rounds to 4,300,000 to the nearest 100,000. What is the greatest possible number?
Answer 4,349,999.
3. Is 2,650,000 rounded to the nearest million 3,000,000?
Answer Yes.
4. Miro says: Miro includes 85,499 in the range that rounds to 86,000. Is this correct? Explain.
Answer 85,499 is nearer 85,000. The lower boundary is 85,500.

Reason about rounding boundaries

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. A number rounds to 4,300,000 to the nearest 100,000. What is the greatest possible number?

2. Is 2,650,000 rounded to the nearest million 3,000,000?

3. Two different numbers both round to 720,000 to the nearest 10,000. What is the greatest possible difference?

4. Identify and correct this misconception: Miro includes 85,499 in the range that rounds to 86,000.

5. Write a complete sentence explaining how you checked one answer.

Teacher answers and checking prompts

1. A number rounds to 4,300,000 to the nearest 100,000. What is the greatest possible number?
Answer 4,349,999.
2. Is 2,650,000 rounded to the nearest million 3,000,000?
Answer Yes.
3. Two different numbers both round to 720,000 to the nearest 10,000. What is the greatest possible difference?
Answer 9,999, using 715,000 and 724,999.
4. Identify and correct this misconception: Miro includes 85,499 in the range that rounds to 86,000.
Answer 85,499 is nearer 85,000. The lower boundary is 85,500.
5. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.

Reason about rounding boundaries

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Two different numbers both round to 720,000 to the nearest 10,000. What is the greatest possible difference?

2. Explain why this reasoning fails: Miro includes 85,499 in the range that rounds to 86,000.

3. Create a new example that tests the same idea as: Is 2,650,000 rounded to the nearest million 3,000,000?

4. Find a second method or representation. Compare its efficiency with your first method.

5. Write one always, sometimes or never statement about today's learning and justify it.

Teacher answers and checking prompts

1. Two different numbers both round to 720,000 to the nearest 10,000. What is the greatest possible difference?

Answer 9,999, using 715,000 and 724,999.

2. Explain why this reasoning fails: Miro includes 85,499 in the range that rounds to 86,000.

Answer 85,499 is nearer 85,000. The lower boundary is 85,500.

3. Create a new example that tests the same idea as: Is 2,650,000 rounded to the nearest million 3,000,000?

Answer Answers vary. The example and solution must preserve the mathematical structure.

4. Find a second method or representation. Compare its efficiency with your first method.

Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.

5. Write one always, sometimes or never statement about today's learning and justify it.

Answer Answers vary. A valid example and counterexample should support the classification.